Software Engineering – Assessment Tool 1

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Title: Intelligent Nuclear Power Plant Monitoring Platform

Industry Context

Nuclear power plants generate electricity by operating nuclear reactors under highly controlled conditions. Continuous monitoring of reactor temperature, pressure, coolant flow, radiation levels, and equipment health is essential to ensure safe and efficient operation. Traditional monitoring methods require significant human supervision and may not identify failures at an early stage. An intelligent monitoring platform can collect real-time data from sensors, analyze operational conditions, detect abnormal situations, predict equipment failures using Artificial Intelligence (AI), and provide emergency alerts to plant operators. This improves plant safety, reliability, and regulatory compliance.

Problem Statement

Design and develop an Intelligent Nuclear Power Plant Monitoring Platform that continuously collects real-time data from reactor sensors, radiation detectors, cooling systems, turbines, and safety equipment. The platform should monitor reactor conditions, detect operational anomalies, predict equipment failures using machine learning, monitor radiation levels, support emergency response, and generate compliance reports for regulatory authorities. The system should provide centralized dashboards, instant notifications, historical reports, and secure access for authorized personnel.

Objectives

1. Monitor reactor conditions continuously.
2. Detect operational anomalies in real time.
3. Predict equipment failures before breakdown.
4. Track radiation levels across the plant.
5. Support emergency response and alert management.

6. Generate compliance analytics and safety reports.

Expected Outcomes

1. Enhanced plant safety.
2. Reduced operational risks.
3. Improved regulatory compliance.
4. Lower unplanned equipment downtime.
5. Better decision support for plant operators.

Proposed Solution

The proposed system is a cloud-enabled intelligent monitoring platform that gathers data from IoT sensors installed throughout the nuclear power plant. AI algorithms analyse collected data to detect abnormal operating conditions and predict failures before they occur. The system immediately alerts operators through dashboards, SMS, email, or mobile notifications and stores historical data for reporting and maintenance planning.

Functional Requirements

- User authentication
- Reactor monitoring
- Radiation monitoring
- Equipment health monitoring
- Failure prediction
- Emergency alerts
- Dashboard
- Compliance reporting
- Historical records

Non-Functional Requirements

Performance: Real-time monitoring
Reliability: 24×7 operation
Security: Encrypted access and role-based authentication
Scalability: Multiple reactors
Availability: High availability
Maintainability: Easy updates

Modules

1. Reactor Monitoring
2. Radiation Monitoring
3. Equipment Health Monitoring
4. AI Failure Prediction
5. Emergency Response
6. Analytics & Reporting

Users

• Plant Operator
• Maintenance Engineer
• Safety Officer
• Plant Administrator

Inputs

Reactor temperature, pressure, coolant flow, radiation readings, equipment sensor data, turbine status, pump status, generator status.

Outputs

Live dashboard, radiation reports, equipment health reports, failure alerts, emergency notifications, compliance analytics, historical reports.

Technology Stack

Frontend: HTML, CSS, JavaScript
Backend: Java/Python/Node.js
Database: MySQL/PostgreSQL
Cloud: AWS/Azure
AI: Machine Learning
IoT: Sensor Network

Advantages

Improves plant safety, detects failures early, reduces maintenance cost, minimises downtime, supports compliance, improves decision making.

Future Enhancements

Digital Twin, Deep Learning-based prediction, drone inspection, remote monitoring app, national grid integration.

Conclusion

The Intelligent Nuclear Power Plant Monitoring Platform improves the safety, reliability, and efficiency of nuclear power plants by combining IoT, AI, cloud computing, and real-time analytics. It reduces operational risks, predicts failures early, improves compliance, and supports better operational decisions.